

On the Selection of Covariates for Transportability: A Doubly Robust ML Approach for Generalizing a Food Labelling Trial

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The Core Problem: The Generalizability Gap

- Randomized Controlled Trials (RCTs) are the gold standard for causal inference but often lack external validity.
- Participants in trials are rarely a random sample of the target population, leading to ****covariate shift****.
- **Our Case Study:** We see a significant covariate shift between our food label RCT and the population of the UK's National Diet Nutrition Survey (NDNS).

Key Imbalance Example

The trial sample is more educated (87% high vs. 68%), more actively trying to lose weight (54% vs. 18%), and has a different age and ethnic distribution than the target population.

Covariate selection is crucial in transportability methods to ensure the validity of our conclusions.

The Methodological Challenge: Covariate Selection

- Transportability methods aim to correct for this covariate shift, but are sensitive to the choice of adjustment covariates.
- This creates a critical ****bias-variance trade-off****, formalized by Colnet et al. (2025):

Bias Reduction

Must adjust for all covariates that are:

- Treatment effect modifiers
- AND shifted between populations

Omitting them leads to a biased PATE.

Variance Inflation

Adjusting for covariates that are:

- Only predictive of trial selection
- OR only predictive of the outcome

...inflates the variance of the PATE estimator.

Our Objective: To empirically investigate this trade-off in a real-world setting with mixed-type (continuous & categorical) data, which is a noted

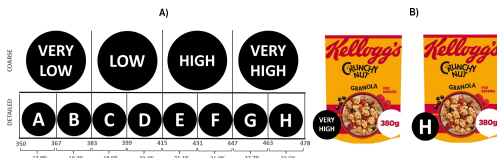
Our Approach: Data and Methodology

- ➊ **Case Study:** A food label RCT where label effectiveness is theoretically moderated by a **continuous effect modifier** (working memory/cognition), alongside standard **shifted demographic covariates**.
- ➋ **Pragmatic Data Solution:** Cognition was missing in the target population (NDNS). We developed a transparent, semi-synthetic data generation process (documented in appendix) to impute this key variable.
- ➌ **Flexible Modelling:** We employ a suite of estimators, culminating in a **Doubly Robust (DR) BART** model to flexibly handle the complex, non-linear relationships in our mixed-type data and propagate model uncertainty.
- ➍ **Systematic Comparison:** We test four estimators and three theoretically-motivated covariate sets (Demographics, Extended, Parsimonious) to directly assess the impact of covariate selection.

The Case Study: A Food Label Granularity RCT

Grounded in Cognitive-Load Theory

- **Objective:** To test if the effectiveness of FOP labels depends on the interaction between label **granularity** and consumer's **working memory capacity**.
- **Outcome:** Calorie count of the cereals selected in the trial.
- **Design:** 3-arm online RCT (N=498) with participants assigned to:
 - 1 No Label (Control) = a
 - 2 Coarse Label (4 categories) = c
 - 3 Detailed Label (8 categories) =
- **Justification:** This design provides a perfect real-world test case for transportability.
 - It has a powerful, continuous **effect modifier** (working memory).
 - It has standard **shifted covariates** (demographics, health behaviours).



Key Estimators and Theoretical Framework

Target Estimand: Population Average Treatment Effect (PATE)

The goal is to estimate the average treatment effect in the target population ($S = 0$).

$$PATE(a, a') = \mathbb{E}[C(a) - C(a') | S = 0]$$

Inverse probability sampling weight (IPSW) Estimator

This is the primary tool for re-weighting. The stratified form reveals the core of the covariate selection problem:

$$\hat{\tau}_{IPSW} = \sum_{x \in \mathcal{X}} \hat{p}(x | S = 0) \hat{\tau}(x)$$

It calculates the CATE, $\hat{\tau}(x)$, within strata from the trial, then weights them by the prevalence of those strata in the target population, $\hat{p}(x | S = 0)$.

The Doubly Robust (DR-BART) Estimator

The DR estimator: direct outcome modeling and inverse probability weighting.

The DR Formula for a Mean Potential Outcome $\mu(a)$

$$\hat{\mu}_{DR}(a) = \underbrace{\frac{1}{m} \sum_{j: S_j=0} \hat{\mu}(a, X_j)}_{\text{Direct Method Term}} + \underbrace{\frac{1}{n} \sum_{i: S_i=1} \frac{\mathbf{1}\{L_i = a\}}{p(L_i = a)} \frac{1 - \hat{\pi}(X_i)}{\hat{\pi}(X_i)} (C_i - \hat{\mu}(a, X_i))}_{\text{IPSW Correction Term}}$$

- **Direct Method Term:** An estimate of the outcome for the target population (m individuals), based on the outcome model ($\hat{\mu}$) trained on the RCT.
- **IPSW Correction Term:** Uses the residuals ($C_i - \hat{\mu}$) from the outcome model in the trial (n individuals) and re-weights them using the participation model ($\hat{\pi}$) to correct for bias.

The "Doubly Robust" Property

The final PATE estimate ($\hat{\tau}_{DR} = \hat{\mu}_{DR}(a) - \hat{\mu}_{DR}(a')$) is consistent if **either** the outcome model ($\hat{\mu}$) **or** the participation model ($\hat{\pi}$) is correctly specified. This provides strong protection against model misspecification.

Covariates Used for Transportability Adjustment

A theoretically-driven set of 9 variables common to both datasets

Demographic

- Age
- Sex
- Ethnicity
- Country

Health-Related

- Body Mass Index (BMI)
- Trying to Lose Weight

Cognitive (Effect Modifier)

- Working Memory (dprime3)

Socioeconomic

- Education Level
- Children in Household

Note: The Working Memory (dprime3) score was measured in the RCT and synthetically generated for the NDNS population.